```
        if not looks_like_header:
            ws.insert_rows(1)
        for idx, value in enumerate(headers, start=1):
            ws.cell(row=1, column=idx, value=value)

    def _to_icao_excel_record(self, row):
        if not isinstance(row, dict):
            return None

        icao_raw = (
            row.get("icaoAddress")
            or row.get("icao")
            or row.get("hex")
            or ""
        )
        icao = str(icao_raw).strip().upper()
        callsign = str(row.get("callsign", "")).strip().upper()

        if not icao or icao == "UNKNOWN":
            if not callsign:
                return None
            # No decoded Mode-S ICAO - identify this flight by its callsign
            # so it still gets logged to its own excel file/sheet.
            identifier = "CS_" + callsign
        else:
            identifier = icao

        safe_icao = "".join(ch for ch in identifier if ch.isalnum())
        if not safe_icao:
            return None

        ts_iso, _ = _parse_ts(row.get("timestamp"))
        callsign = callsign or "UNKNOWN"

        lat = round(_to_float(row.get("latDD", row.get("lat", row.get("latitude", 0.0))), 0.0), 8)
        lon = round(_to_float(row.get("lonDD", row.get("lon", row.get("longitude", 0.0))), 0.0), 8)

        altitude_mm = _to_float(row.get("altitudeMM"), None)
        altitude_m = (
            round(altitude_mm / 1000.0, 3)
            if altitude_mm is not None
            else _to_float(row.get("altitude"), 0.0)
        )
        heading_de2 = _to_float(row.get("headingDE2"), None)
        heading_deg = (
            round(heading_de2 / 100.0, 2)
            if heading_de2 is not None
            else _to_float(row.get("heading"), 0.0)
        )
        speed_cms = _to_float(row.get("horVelocityCMS"), None)
        speed_mps = (
            round(speed_cms / 100.0, 3)
            if speed_cms is not None
            else _to_float(row.get("speed"), 0.0)
        )
        vs_cms = _to_float(row.get("verVelocityCMS"), None)
        vertical_rate_mps = (
            round(vs_cms / 100.0, 3)
            if vs_cms is not None
            else _to_float(row.get("verticalRate"), 0.0)
        )

        return {
            "timestamp": ts_iso,
            "icao": safe_icao,
            "callsign": callsign,
            "latitude": lat,
            "longitude": lon,
            "altitude": altitude_m,
            "heading": heading_deg,
            "speed": speed_mps,
            "vertical_rate": vertical_rate_mps,
        }
```